Beyond the Combined SAT Proposal Frontier

Multi-stage contraction schedules, faster exact labeling,
and residual-precision arithmetic

Research prepared for Vladimir Reshetnikov
with ChatGPT

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Abstract

We review the seven filed refinement packets in ProveIt's SubexponentialSAT project, eight second-round packets from `docs/incoming`, and their two primary manuscripts. The correct comparison includes compatible compositions of the newer packing and Cauchy-mode improvements. We give a multi-stage polynomial-space contraction schedule with coefficient $63.8522033905\ldots$ on the scale $s/\sqrt{\log_2 s}$, below the composed $66.6051434076\ldots$ certificate and a strengthened two-stage comparison near 66.1205. A separately optimized cached construction gives time coefficient $37.9301090829\ldots$ and space coefficient $33.6230409217\ldots$, below the fixed-alphabet composed-family time infimum $38.1677456970\ldots$.

Two executable changes complement these mathematical certificates. Caching row sums in the existing exact packet-labeling implementation reduces its candidate-scoring phase from $O(NQ^3)$ to $O(NQ^2)$ rational operations, preserving every score and tie break. Complete certificates agree on 840 cases; the largest of five benchmark fixtures improves from about 304 ms to 32 ms. An exact valuation-normalization identity moves suitable mode arithmetic from P bits to $P - V(\beta)$ bits, with an explicit correction to the creation-precision invariant. Its prototypes pass independent actual-domain Cauchy checks and show useful arithmetic-kernel savings.

The global SAT logarithmic orders remain unchanged. Numerical coefficients are per normalized tensor wire, not per SAT variable. The contraction results retain stated upstream algebraic dependencies; no full SAT solver, new Lean/Rocq certificate, or end-to-end SAT speedup is claimed.

Review pin: `474ddba696a5d385c091c23fb2995f69edd13e00`.
Closing placement reviewed: `6d3adc45061535ebe6fa7cd3d9b37dff7dcfc22a`, October 26, 7:03 pm Pacific.
Evidence: source hashes, exact checkers, raw benchmark trials, reference code, and complete editable report sources accompany this PDF.

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1 Results and the comparison standard

The useful outcomes are an actual improvement to a supplied implementation and smaller explicit contraction certificates after combining the relevant prior ideas. Merely comparing with the first-round coefficient 648.44 would miss most of the current frontier.

Let s be the normalized Boolean wire bound, and write

$$\mathfrak{E}(s) = \frac{s}{\sqrt{\log_2 s}}.$$

Suppress fixed polynomial factors in s and the tensor-description length $\mathcal{L}$. Table 1 distinguishes polynomial-space enumeration from the cached algorithm, which spends subexponential space. The new certificates are proved in Section 4.

Table 1: Comparable leading exponential certificates. Every time entry means $2^{(C+o(1))\mathfrak{E}(s)}$ up to a fixed polynomial. A numerical space entry has the same interpretation.

Construction or derived comparison	Time C	Space
Best displayed incoming polynomial-space schedule	76.9090	polynomial
That schedule with full-block slope-four modes	66.6051	polynomial
Strengthened fixed-16 boundary comparison	66.1927	polynomial
Derived root/tail comparison (17, then 13)	66.1205	polynomial
New multi-stage schedule	**63.8522**	**polynomial**
Incoming asymmetric schedule with full-block caching	44.5376	39.4959
Retuned fixed-8 composed-family boundary	38.1677	33.9927
New root-7/tail-9 cached schedule	**37.9301**	**33.6230**

Boundary comparisons are limiting parameter certificates, not empirical measurements or lower bounds on what the earlier ideas could ever achieve. The root/tail polynomial-space comparison is an additional compatible construction derived during this review. None of these numerical searches is used as a proof of global parameter optimality. The new schedules use explicit rational parameters and strict packing margins.

The implementation improvement in Section 5 has a different evidentiary status: it is benchmarked against a byte-identical existing implementation and returns the same complete results. Section 6 gives a second exact transformation and separately scoped arithmetic-kernel benchmarks. The ratios from these experiments must not be multiplied by exponent-coefficient ratios to predict a full SAT speedup.

1.1 The exact input interface

The contraction theorems use the interface common to the newer packets. For fixed constants A, b, C, the input is a bipartite Boolean tensor network with at most s edges, polynomially many vertices, maximum degree at most $\left\lceil s^{9/10} \right\rceil$, and total description length $\mathcal{L}$. Parallel edges are allowed and have distinct ports. Entries belong to

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \le P_0, \ell \le Cs.$$

Every original entry is evaluable in time $(s+\mathcal{L})^b 2^{As^{9/10}}$ and polynomial space. The extension-ring arithmetic, deformation, recovery, and recursive factorization are those of the cited preprint [3],

with the refinements explicitly identified below. The constants in the $o(1)$ terms are uniform for fixed interface constants. Bounded exceptional sizes can be handled by direct summation.

Theorem 1.1 (The new polynomial-space certificate)**.** *Under this interface and the imported identities stated in Section 3, there is a deterministic contraction construction using polynomial space and time*

$$(s+\mathcal{L})^{O(1)}2^{(63.8522033905488\ldots+o(1))\mathfrak{E}(s)}.$$

In particular, coefficient 64 *is valid for all sufficiently large* s*.*

Theorem 1.2 (The new cached certificate)**.** *Under the same interface and the imported cached-factorization machinery, there is a deterministic construction with*

$$T(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)}2^{(37.9301090829\ldots+o(1))\mathfrak{E}(s)},$$
$$S(s,\mathcal{L}) \le (s+\mathcal{L})^{O(1)}2^{(33.6230409217\ldots+o(1))\mathfrak{E}(s)}.$$

Thus coefficients 38 *and* 34 *respectively are valid eventually.*

These are ordinary mathematical derivations relative to stated imported lemmas, with independent component review and exact parameter checks. They are not independently formalized versions of the entire ETH claim. Their small exponent margins also imply potentially enormous crossover sizes. The finite timing evidence concerns the two implemented kernels, not these asymptotic crossovers.

1.2 What "all proposals" can mean rigorously

The review compares results on the same input and resource interface and composes compatible improvements before proposing another one. In particular, the asymmetric packet already counts the fraction $1-1/r$ of unequal symbols; applying the round-two crossing saving again would be incorrect. The full-block slope-four basis replaces the older slope-three basis without changing its family multiplicity. The cached cost must be re-derived with the new shrinkage and taper schedule rather than scaled by a ratio taken from a different schedule.

There are two necessary limits to an all-proposals numerical comparison. The separate Claude manuscript leaves its relevant leading constant unspecified, so a numerical dominance statement against it is unavailable [4]. Several first-round packets state faster logarithmic orders only under unconstructed algorithmic hypotheses. A finite improvement of the established $s/\sqrt{\log s}$ coefficient cannot dominate an assumed $s/\log s$ algorithm as an asymptotic function. We retain those as open targets. Low-degree specializations are also retained on their stronger domains rather than compared as if they had the general degree interface.

A portfolio of the applicable established constructions and the new ones preserves the previous pointwise upper envelopes. For each fixed resource class, a finite dovetailing has constant-factor scheduling overhead; alternatively one can choose by their explicit input-dependent guards. The strict new improvements asserted here concern the displayed common interface and the actual packet-labeling code. We do not claim to have optimized every possible recombination of all existing lemmas.

2 Repository review and provenance

The closing read-only checkout is pinned to `474ddba696a5d385c091c23fb2995f69edd13e00`. The initial review used `ab2fbc9e03d5c3d0ad6dec8686960bfe498b0b06`, including the seven ZIPs ar-

riving in `3e57705c`. The closing check added the matching packet from `d098ba43` and reviewed placement commit `6d3adc45`, which filed all eight second-round packets under `Research`. The earlier commits are `45233238` (summary of claimed improvements), `3e096089` (vendored primary preprint), `2b925d88` (placement of the first seven packets), and the arrivals `8afdbbea`, `64b3202e`, and `a505bd76` [1, 2]. A byte comparison confirms all 187 retained relocated files from the original seven second-round archives are unchanged; the only additional packet has 24 filed files. The manifest retains initial and closing identifiers, per-file hashes, and archive members. No remote repository file was modified.

The repository README explicitly treats the material as external research and reports no checked ProveIt Lean/Rocq development for this project [1]. The current work reads the original reports and relevant source code rather than treating that README as a proof certificate. The finite verification performed here applies to the delivered changes; it does not retrospectively certify every upstream claim.

2.1 Seven established packets

Packet	Result relevant to the strongest comparison
Factorial Cauchy modes [5]	Integral features with weights $1 + j + 2v_2(j!) = 1 + 3j - 2s_2(j)$; leading mode coefficient divided by $\sqrt{3}$. The row offset was already observed. This is a count/certificate improvement.
Contraction improvements [6]	Exact taper count, working precision near $6MD/u$, schedule coefficient 648.4411, and streamed coefficient extraction. No full solver timing.
Precision refinement [7]	Exact recovery loss and the full-loss taper minimax problem. The optimal leading scalar loss remains $6M/u$. A stronger structured-summation algorithm is assumed rather than constructed.
Uniform density improvement [8]	Uniform density-sensitive exponent $\frac{n}{\sqrt{\log n}} \min\{\log(2 + m/n), \log\log n\}$. Expander and contraction theorems remain imported dependencies.
Density refinement [9]	Overlapping density improvement and precision constants. Contains an earlier local odd-unit Cauchy Smith profile; that profile is not new here.
Scalar parity refinement [10]	Exact scalar parity lift, reducing an incidence-degree cap to $k(p + a - 1)$. Faster recursion still needs a missing compact factorization.
Asymptotic analysis [11]	Explicit barriers for the analyzed mode/taper mechanisms and two missing interfaces. No stronger unrestricted bound is established.

The strongest conditional target among these packets is $2^{O_k(n \log\log n/\log n)}$, with $n/\log n$ at fixed density [7]. A smaller index encoding alone does not provide the required cheap summation or bounded-wire group tensors. The exact taper minimax result already excludes a further order improvement from merely retuning that taper. The present work does not rebrand either observation as a new result.

2.2 Eight second-round packets, including the closing arrival

Archive family	Contribution and displayed coefficient
Asymmetric balancing [12]	Nearly unit padding, separate side degrees, exact crossing fraction, additive precision buffer; 76.9090 on the full interface. Its low-degree specialization has stronger hypotheses. Its annulus slopes tend to three.
Balanced packing [13]	Deterministic nearly uniform pair loads and capacity scheduling; 94.7482. Symmetric degree treatment gives less cumulative capacity.
High-moment packing [14]	Matching/dependency control and an even-moment construction; 94.7482. Its $\exp(O(\log^2 s))$ setup is paid and is lower order on the principal scale.
Matching refinement [19]	The closing arrival gives another symmetric matching-potential derivation of 94.7482, with fixed alphabet 54 and an adaptive thin-gap schedule. It does not alter the combined frontier.
Recursive refinement [15]	A two-level schedule and stronger reciprocal-domain Cauchy weights, with uniform slope 64/21; 133.2074 as displayed. The full-block slope-four basis is the stronger compatible local ingredient.
Round-two refinement [16]	Matching-potential packing, exact crossing fraction, thin precision gap, and different root/tail alphabets; 84.0663. Changing alphabets by depth is therefore already a prior idea.
Packet refinement [17]	Two-stage pair balancing, near-unit padding, rational schedule, and additive precision guard; 95.6639.
Joint Cauchy modes [18]	Group-preserving full-orientation slope-four features at every valuation; 319.6858 in its old polynomial-space schedule. Optional cached group tables and Strassen aggregation give 225.5971 time and 196.1994 space on that schedule. These ingredients must be recomposed with better packing.

The archive names and exact member paths appear in the source manifest. The review establishes novelty only relative to this checked corpus. Classical conditional expectation, Cauchy separation, and Strassen multiplication are not claimed as inventions of this report. The new theoretical contribution is the particular capacity-aware multi-stage and cache-aware schedules, including their complete resource analysis; the new executable contributions are the two transformations proved below.

3 Inherited estimates and their compatibility

Let $\lambda = \lceil \log_2 s \rceil$ and let an active call have $M > s/\lambda$ wires. Its packet length is $L = 2^a$, its alphabet is r, its whole-vertex label count is $Q = r^a$, and at an active interpolation node of valuation $i < u$ put $D = 2^{a-i}$. In this section P denotes arithmetic precision; we use B for the common padded wire mass.

3.1 Asymmetric padding and child sizes

The asymmetric two-stage labeling gives $\mathsf{B} = (1 + o(1))M$ whenever

$$QR/M = o(1), \qquad Q^2C\lambda^2/M = o(1), \qquad LQ^2/M = o(1), \tag{1}$$

where R, C bound degrees on the two sides [12]. Its explicit error under our fixed power gaps is $\delta_s = O(1/\lambda) + s^{-c+o(1)}$ for a fixed $c > 0$. Thus $\delta_s \log \lambda = o(1)$, which controls accumulated padding through the recursion; a separate unspecified $o(1)$ at each level would not be enough to justify a leading coefficient.

At the root both sides have degree at most $\Delta = \left\lceil s^{9/10} \right\rceil$. At every later call put retained tensors on the row side and fresh packet tensors on the column side. Then

$$R_d \le \Delta, \qquad C_d \le 2L_{d-1}. \tag{2}$$

Every child has at most $2\mathsf{B}/r$ wires. For valuation i, unequal symbols account for exactly the fraction $1 - 1/r$ of common padded cells, giving at most

$$c_i \le (2(1 - 1/r) + o(1))\, M/D \tag{3}$$

families of any one mode index. For a specified directed pair of different groups the sharper count is

$$c_{gh} \le (2 + o(1))M/(r^2 D). \tag{4}$$

These counts follow from the uniform padded cell arrangement, not from an independence assumption about the realized labels.

3.2 Sharp precision with a logarithmic buffer

The exact Hermite condition count for $q_i = h(u - i)$ is $hL(2u - 2 + 2^{1-u})$. Taking

$$h = \left\lceil \frac{M(2L-1)+1}{L(2u-2+2^{1-u})} \right\rceil$$

gives enough conditions for deformation degree $M(2L - 1)$. The established safe working allowance is

$$P_i = p + 6hD + hu(a + 1), \tag{5}$$

where p is inherited precision [6, 7]. Use

$$b = 6\left\lceil \log_2 \lambda \right\rceil, \qquad a_d = \sum_{j=0}^{d} u_j + (d + 1)b, \tag{6}$$

and $a_{-1} = 0$. At logarithmic depth, inherited precision satisfies $p = O(s\lambda^2 2^{a_{d-1}})$. Since $i < u_d$, $D \ge 2^{a_{d-1}+b+1}$, whence

$$\frac{p}{MD/u_d} = O(\lambda^4 2^{-b}) = o(1).$$

The last jet-evaluation term in (5) is also lower order. Once $u_d \to \infty$ uniformly, $h = (1+o(1))M/u_d$, so

$$P_i \le (6 + o(1))MD/u_d. \tag{7}$$

The schedules below prove that required uniform divergence rather than assuming it from a fixed-depth calculation.

3.3 Full-block modes and entropy

The joint-Cauchy packet proves a finite, group-preserving separation with integral source features and target weights

$$w_j = a + 1 - i + 4j - 2s_2(j), \qquad 0 \le j < D/2, \tag{8}$$

where s_2 is binary popcount [18]. It works over the source's unramified extension, including non-prime residue classes, and retains the family multiplicity in (3). Its normalized triangular Cauchy factors and their Smith valuations are prior results, not newly proved here.

Conservatively lower the offset in (8) to one and extend the mode index range. The generating product gives

$$\sum_{j\ge 0} -\ln(1 - e^{-\theta(1+4j-2s_2(j))}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

Combining (3) and (7) and optimizing the exponential tilt yields

$$\log_2 N_i \le (K(r) + o(1))\frac{M}{\sqrt{u_d}}, \qquad K(r) = \frac{\pi\sqrt{2(1-1/r)}}{\ln 2}. \tag{9}$$

The saddle has $\theta \asymp \sqrt{u}/D \to 0$ uniformly. In particular, the error relative to the leading bound is $O((\sqrt{u}/D)\log^2(D/\sqrt{u})) = o(1)$, since $D \ge \lambda^6$ and $u = O(\lambda)$; the ratio is decreasing for sufficiently large $D/\sqrt{u}$. All alphabets used below lie in a fixed finite set.

The source's child topology and immutable entry descriptors do not require the alphabet to be the same at consecutive depths. Labels are freshly chosen for the current graph. What carries history is the cumulative packet exponent and the entry creation convention. This validates the multi-stage compositions while preserving the existing factorization.

4 New explicit contraction schedules

4.1 Polynomial space: use the side-degree capacity at several depths

The root pays the symmetric high-degree constraint $Q_0^2\Delta \ll s$, while subsequent calls need only $Q_d\Delta \ll s$ for their old side. Their fresh side is much smaller by (2). The asymmetric packet already exploits this distinction, and the round-two packet already changes the root alphabet. Here the cumulative capacity is used at several depths, with progressively smaller tail alphabets.

Choose the following rational retained-range coefficients:

d	0	1	2	3	$4+k$ $(k \ge 0)$
r_d	13	14	8	6	4
η_d	27/2000	51/4000	141/20000	107/20000	$(27/6400)(5/8)^k$.

(10)

Use $u_d = \lfloor \eta_d \lambda \rfloor$, the buffer (6), and leaf threshold $H = \lceil s/\lambda \rceil$. All parameters are computable by integer arithmetic. The cumulative leading ranges are

$$A_0 = \frac{27}{2000}, \quad A_1 = \frac{21}{800}, \quad A_2 = \frac{333}{10000}, \quad A_3 = \frac{773}{20000}, \quad A_\infty = \frac{499}{10000}. \tag{11}$$

Lemma 4.1 (Every packing gate has a strict margin)**.** *At the root, $2A_0 \log_2 r_0 < 1/10$. At all later depths, $A_d \log_2 r_d < 1/10$. The three conditions in (1) hold uniformly at active logarithmic depths, and all new tensor degrees stay below Δ.*

Proof. The required scalar inequalities have exact integer certificates

$$13^{27} < 2^{100}, \quad 14^{21} < 2^{80}, \quad 999 < 1000, \quad 6^{773} < 2^{2000}, \quad 998 < 1000.$$

They correspond in order to the root and the four tail alphabet regimes. At the root they give $Q_0^2\Delta = s^{1-c_0+o(1)}$ for fixed $c_0 > 0$. Below the root they give $Q_d R_d = s^{1-c_d+o(1)}$ with a uniform positive minimum margin. The fresh-side exponent satisfies

$$2A_d \log_2 r_d + A_{d-1} < \frac{1}{5} + \frac{499}{10000} = \frac{2499}{10000} < 1.$$

The same bound controls $L_d Q_d^2$. The added buffers cost only $O((\log\lambda)^2) = o(\lambda)$ at logarithmic depth, while active calls have $s/M < \lambda$. These logarithmic factors are absorbed by the strict power margins. Finally $2L_d = s^{0.0499+o(1)} < \Delta$. □

Noncircular depth argument. First prove the preceding estimates for every hypothetical active depth

$$d \le \left\lceil \log_{5/3} \lambda \right\rceil + C_0$$

with a fixed sufficiently large C_0. In that range,

$$u_d \ge \lambda^{\epsilon_0+o(1)}, \qquad \epsilon_0 = 1 - \frac{\ln(8/5)}{\ln(5/3)} > 0,$$

where positivity follows from $24 < 25$. Thus floors are uniformly negligible, all taper ranges diverge, and the packing and precision estimates are valid throughout the hypothetical range. For sufficiently large s, every child has at most $3M/5$ wires, because $r_d \ge 4$ and padding tends uniformly to one. No active path can then pass the stated depth. This establishes the depth bound without assuming the desired shrinkage before verifying packing.

Using the sharper accumulated padding estimate gives

$$M_d \le (1+o(1)) s f_d, \quad (f_0, f_1, f_2, f_3) = \left(1, \frac{2}{13}, \frac{2}{91}, \frac{1}{182}\right), \quad f_{4+k} = \frac{1}{546}\, 2^{-k}. \tag{12}$$

Apply (9) and sum the logarithms of branching counts along a path. The tail work ratio is $(1/2)/\sqrt{5/8} = \sqrt{2/5} < 1$. Therefore the exact leading coefficient is

$$\begin{aligned} C_{\mathrm{ps}} = {} & \frac{K(13)}{\sqrt{27/2000}} + \frac{2K(14)}{13\sqrt{51/4000}} + \frac{2K(8)}{91\sqrt{141/20000}} \\ & + \frac{K(6)}{182\sqrt{107/20000}} + \frac{K(4)}{546\sqrt{27/6400}\,(1-\sqrt{2/5})}. \end{aligned} \tag{13}$$

The exact rational interval checker proves

$$63.85220339054 < C_{\mathrm{ps}} < 63.85220339056 < 64. \tag{14}$$

It uses rational bounds from Machin's arctangent formula for π, a positive series for $\ln 2$, and integer-square-root brackets. A floating-point parameter search suggested the schedule but is not an input to the proof.

Polynomial choices of interpolation node, residue-field evaluation, and child group over $O(\log\lambda)$ levels contribute $O(\lambda \log\lambda)$ to the logarithm of total work. Leaf summation costs $2^{O(s/\lambda)}$. Original entry oracles cost $2^{O(s^{9/10})}$; new packet entries cost at most $\exp(O(L_d \log s)) = \exp(s^{0.0499+o(1)})$. All are lower order than $\mathfrak{E}(s)$. Polynomial-bit descriptions, arithmetic and precision, together with depth-first traversal, preserve polynomial space. The description-length factor stays a fixed polynomial because branching depends on s, not on $\mathcal{L}$. This proves Theorem 1.1.

4.2 A stronger comparison than the displayed prior coefficients

The incoming asymmetric schedule uses $r = 16$, root range $\lambda/81$, and a geometric tail of the same total range. Composing it with full-block modes multiplies its slope-three coefficient by $\sqrt{3}/2$, giving $66.6051434076\ldots$. Its fixed-alphabet capacity boundary and optimal tail allocation give the stronger derived value

$$K(r)\sqrt{20\log_2 r}\left[1+\frac{2/r}{(1-(2/r)^{2/3})^{3/2}}\right], \tag{15}$$

which at $r = 16$ is $66.1927236936\ldots$.

For a root alphabet r_0 and a constant tail alphabet r_1, set

$$x_0 = \frac{.05}{\log_2 r_0}, \qquad U = \frac{.1}{\log_2 r_1} - x_0 > 0.$$

The corresponding optimally allocated geometric tail has coefficient

$$\frac{K(r_0)}{\sqrt{x_0}} + \frac{(2/r_0)K(r_1)}{\sqrt{U}\,(1-(2/r_1)^{2/3})^{3/2}}.$$

At $(r_0, r_1) = (17, 13)$ it is about 66.12047546. These formulas give feasible limiting comparisons by introducing a vanishing reserve in the capacities; they do not assert global minima over all schedules. The new strict rational coefficient in (14) is below all three comparisons. Its gain is about 4.13% against the direct composed certificate and 3.43% against the stronger root/tail comparison.

4.3 Cached contraction: pay for the group functions

The cached method is inherited from the joint-Cauchy packet [18]. Partition a mode tuple by its exact total weight on every directed pair of groups. With fixed r and polynomial precision, there are only polynomially many weight arrays, although the polynomial degree can be large. For one array, let h_e be the base-two logarithm of the number of exact-weight tuples on pair e. Polynomial-size dynamic programming over modes and weights supplies exact counts and rank/unrank operations; an efficient hidden oracle is not assumed.

Put

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2}, \qquad \mathcal{K}(M,u) = \kappa\frac{M}{\sqrt{u}}. \tag{16}$$

From (4), the slope-four generating function, and the global weight budget, the common-tilt argument gives for fixed nonnegative pair weights z_e,

$$\sum_e z_e h_e \le \left(\sqrt{\frac{\sum_e z_e^2}{r^2}} + o(1)\right)\mathcal{K}(M,u). \tag{17}$$

Using one common tilt matters: it includes zero and small individual weight budgets without incorrectly applying a separate large-budget asymptotic to each pair.

Cache each group's function of its incident pair indices, obtaining each value by a genuine recursive child contraction. Partition the groups into three blocks. Sum internal indices to form block boundary tables, then contract the resulting triangle using division-free rectangular Strassen multiplication. For block size k, the numbers of incident and boundary directed pairs are respectively

$$I_r(k) = 2rk - k^2 - k, \qquad J_r(k) = 2k(r-k).$$

If the three inter-block classes have sizes v_1, v_2, v_3, the rectangular matrix exponent follows from

$$H_1 + H_2 + H_3 - (3-\omega)\min(H_1, H_2, H_3), \qquad \omega = \log_2 7.$$

Writing this as the maximum of three weighted sums and using (17) gives the matrix coefficient

$$b_r = \sqrt{\frac{v_1 + v_2 + v_3 - [1-(\omega-2)^2]\min_j v_j}{r^2}}. \tag{18}$$

The cache, block-construction, and memory coefficients are

$$a_r = \frac{\sqrt{2(r-1)}}{r}, \qquad B_r = \max\left\{b_r, \frac{\sqrt{\max_k I_r(k)}}{r}\right\}, \qquad j_r = \frac{\sqrt{\max_k J_r(k)}}{r}. \tag{19}$$

Here B_r must include both block-table construction and matrix work. Omitting the former would be a mistake for the root schedule below.

Up to the already controlled polynomial and lower-order factors, the correct recurrence is

$$T_d \le 2^{a_{r_d}\mathcal{K}_d}\, r_d T_{d+1} + 2^{B_{r_d}\mathcal{K}_d}. \tag{20}$$

Thus caching does not grant free group tensors. The first term pays for all recursive evaluations, and the second pays for the tables and their contraction. The inherited sequential-record implementation, reblocking and sorting show that polynomial overhead suffices in a bit-cost model; an exponential random-access table is not treated as a unit-cost oracle. Strassen uses only additions, subtractions and multiplications, so it is valid in the source's residue ring with nilpotents.

4.4 Why the stronger fixed-alphabet baseline must be included

At the asymmetric packet's displayed $r = 16$ schedule, the new per-pair counts and full-block cached method give time coefficient $44.5375718550\ldots$ and space coefficient $39.4959495337\ldots$. But one can retune the existing fixed-alphabet family. At $r = 8$, root capacity $x_0 = 1/60$, and a tail of the same total capacity, the appropriate recurrence is root dominated for a range of geometric tail allocations. Its limiting time coefficient is

$$C_{8,\partial} = \kappa\sqrt{60}\sqrt{\frac{30 + 12(\omega-2)^2}{64}} = 38.1677456970\ldots, \tag{21}$$

with space coefficient $33.9927010637\ldots$. Vanishing capacity reserve handles the equality boundary. This is the more demanding comparator used here, rather than only the 44.54 displayed-schedule composition.

4.5 The new strict root-7/tail-9 schedule

Use seven groups at the root, partitioned as $(2,2,3)$, and nine groups at every later depth, partitioned as $(3,3,3)$. Set

$$\eta_0 = \frac{89}{5000}, \qquad \eta_d = \frac{1233}{125000}\left(\frac{7}{25}\right)^{d-1} \quad (d \ge 1), \qquad u_d = \lfloor \eta_d \lambda \rfloor. \tag{22}$$

Keep the buffer (6) and $H = \lceil s/\lambda \rceil$. The tail total is $137/10000$, and the total cumulative capacity is

$$A_\infty = \frac{89}{5000} + \frac{137}{10000} = \frac{63}{2000} = 0.0315.$$

The exact inequalities

$$7^{89} < 2^{250}, \qquad 9^{63} < 2^{200} \tag{23}$$

give the strict root gate $2\eta_0 \log_2 7 < 1/10$ and tail gate $A_\infty \log_2 9 < 1/10$. The fresh-side and padding exponents are less than $1/5 + A_\infty = 0.2315 < 1$. New tensor degrees are at most $s^{0.0315+o(1)} < \Delta$.

Verify these bounds first through hypothetical depth $\lceil \log_4 \lambda \rceil + O(1)$. Tail ranges satisfy

$$u_d \geq \lambda^{1-\ln(25/7)/\ln 4+o(1)} \longrightarrow \infty$$

because $25/7 < 4$. Eventually the root child shrinkage is at most $1/3$ and every tail child shrinkage at most $1/4$, while the sharp factors are $2/7$ and $2/9$ respectively. This closes the same noncircular depth bootstrap as before and validates the uniform precision estimates. Accumulated padding errors give

$$M_0 \leq s, \qquad M_d \leq (1+o(1))s\frac{2}{7}\left(\frac{2}{9}\right)^{d-1} \qquad (d \geq 1).$$

Write $\xi = \omega - 2 = \log_2(7/4)$. At the root,

$$a_7 = \frac{\sqrt{12}}{7}, \quad b_7 = \frac{\sqrt{24+8\xi^2}}{7}, \quad B_7 = \frac{\sqrt{30}}{7}, \quad j_7 = \frac{\sqrt{24}}{7}. \tag{24}$$

Here $b_7 < B_7$: root block construction costs more than root matrix multiplication. In the tail,

$$a_9 = \frac{4}{9}, \quad B_9 = b_9 = \frac{\sqrt{4+2\xi^2}}{3}, \quad j_9 = \frac{2}{3}. \tag{25}$$

The tail block-construction coefficient $\sqrt{42}/9$ is smaller than B_9. The tail work ratio is

$$\gamma = \frac{2/9}{\sqrt{7/25}} = \frac{10}{9\sqrt{7}} < 1. \tag{26}$$

Moreover $a_9/(1-\gamma) = 0.766231\ldots < B_9 = 0.767655\ldots$. This strict inequality ensures that all deeper cache evaluations are covered by the first tail node's local table bound.

Let

$$K_0 = \frac{\kappa}{\sqrt{89/5000}}, \qquad K_1 = \frac{(2/7)\kappa}{\sqrt{1233/125000}}.$$

Unrolling (20), the normalized total time coefficient is at most

$$C_{\text{cache}} = \max\{B_7K_0,\ a_7K_0 + B_9K_1\} = a_7K_0 + B_9K_1. \tag{27}$$

The root table term is $37.59170256\ldots$; the second term, including recursive child work, is $37.9301090829\ldots$ and is the bottleneck. The memory coefficient is

$$C_{\text{mem}} = \max\{j_7K_0, j_9K_1\} = j_7K_0 = 33.6230409217\ldots. \tag{28}$$

Ancestor and descendant table sizes add, so their leading space exponents take a maximum, rather than a sum. There are only logarithmically many stack levels, and the tail $\mathcal{K}_d$ decreases geometrically.

The supplied checker proves these comparisons with rational intervals, including a positive margin in $a_9/(1-\gamma) < B_9$ and the complete root maximum. It encloses

$$37.93010908288 < C_{\text{cache}} < 37.93010908289 < 38,$$
$$33.62304092165 < C_{\text{mem}} < 33.62304092167 < 34.$$

The same lower-order costs already checked for Theorem 1.1 are harmless here, with an even smaller packet exponent. This proves Theorem 1.2. The time coefficient improves by about 0.62% over (21); the comparison is a strict improvement of certificates, not a prediction of a 0.62% runtime reduction.

4.6 Consequences for SAT and the unresolved order improvement

Let n be the declared variable count, m the number of clauses and k a fixed width. The unchanged reduction and the density refinements give the polynomial-space envelope

$$(n+m)^{O_k(1)}\, 2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{29}$$

Small sizes are understood with the usual harmless logarithm offsets. At fixed density this is $2^{O_{k,D}(n/\sqrt{\log n})}$ up to polynomial factors. At unrestricted density it has the $n\log\log n/\sqrt{\log n}$ numerator/denominator profile. The new engine preserves these orders and improves its specified wire-level certificate. Converting it to a numerical coefficient per variable requires the reduction's actual wire and width constants. In particular, writing $2^{64n/\sqrt{\log n}}$ as a universal SAT consequence would be unsupported.

The cached version can use the same reductions at its different space cost. Direct clause-network bounds and prior low-degree specializations remain options in a portfolio. The separate manuscript's unspecified constants cannot be ranked numerically from the supplied information.

The persistent obstacle to a smaller logarithmic order is unchanged: the mode weights still grow linearly, their weighted partitions have the $M/\sqrt{u}$ entropy scale, and packing keeps $u = O(\log s)$. The existing scoped barriers do not prove a lower bound for SAT; they show why the present retuning and exact arithmetic transformations do not establish $n/\log n$ or eliminate the dense reduction's extra $\log\log n$. A further order improvement still requires an actual summation/factorization interface or a different representation, with its oracle and wire costs paid explicitly.

5 A faster implementation of the existing balancing algorithm

The most direct performance result is an optimization of code already in the reviewed repository. It changes neither the mathematical labeling algorithm nor its output. The reference is the asymmetric packet's `code/balanced_packing.py` [12]; a byte-identical copy is included as `code/reference_balanced_packing.py`. The source already caches products involving unassigned columns. Our improvement also caches the row sums used to score every candidate label.

5.1 Exact conditional expectations

After the deterministic greedy row labeling, let Q be the number of labels and let w_{av} be the number of edges from row-label class a to column vertex v. Let X_{ab} be the loads from columns assigned so far. The reference uses the rational base

$$z = 1 + \frac{1}{h\max(1,C)},$$

where C is the maximum column degree and $h \geq 2$ is the smoothness parameter. For each row-label class, define

$$F_a = \prod_{v\ \text{unassigned}} \frac{Q-1+z^{w_{av}}}{Q}, \qquad S_a = \sum_{b=0}^{Q-1} z^{X_{ab}}.$$

The exact conditional expectation is $\Phi = \sum_a F_a S_a$. Before assigning column v, remove its factor:

$$F'_a = F_a \left(\frac{Q-1+z^{w_{av}}}{Q}\right)^{-1}.$$

All these are ordinary positive rational numbers. They are unrelated to the nonunit divisions in the two-adic contraction ring.

Proposition 5.1 (Cached candidate scores)**.** *The conditional expectation after assigning v to label b is exactly*

$$\Phi_b = \underbrace{\sum_a F'_a S_a}_{B} + \sum_a F'_a (z^{w_{av}} - 1) z^{X_{ab}}. \tag{30}$$

After selecting b_*, update

$$S_a \leftarrow S_a + (z^{w_{av}} - 1) z^{X_{ab_*}}, \quad z^{X_{ab_*}} \leftarrow z^{X_{ab_*}} z^{w_{av}}, \quad F_a \leftarrow F'_a.$$

Every score, its minimizing label with the original tie rule, and the conditional-expectation certificate are identical to the reference.

Proof. Only cell (a, b) changes when v receives label b. Its contribution to S_a changes from $z^{X_{ab}}$ to $z^{X_{ab}+w_{av}}$. Subtract the former from the latter and factor the difference to obtain (30). The displayed updates preserve the definitions of S_a, the cell powers, and F_a. Induction over columns proves equality with the reference computation. Since the rational scores are exactly equal, their ordering, all ties, and the selected labels are equal as well. □

For N column vertices, the reference loops over Q candidate labels, Q row classes, and Q column classes per assignment. Its candidate-score phase costs $O(NQ^3)$ rational operations. Formula (30) computes the common base once and uses a Q-term correction for each of Q labels, giving $O(NQ^2)$ rational operations. The additional cell-power cache has Q^2 rational entries. Greedy row assignment, the adjacency scan, and the shared table of powers $z^0, \ldots, z^M$ remain unchanged. For example, with R_0 input row vertices, a direct overall arithmetic count is

$$O(M + R_0 N + R_0 Q + (N+1)Q^2)$$

for the new implementation, with $NQ^3 + Q^2$ in place of $(N+1)Q^2$ for the reference's scoring and cell bookkeeping. These are rational-operation counts; the cost of multiplying long integer numerators and denominators must still be paid. All operands have the same polynomial bit-length envelope as in the original method. We do not assert a factor-Q bit-time speedup on every input.

5.2 Comparison with the actual reference code

The verification program compares complete returned objects, including every exact potential, selected label, cell load, and certificate string. It passed 840 full-certificate comparisons, covering 9,724 candidate scores; 840 additional comparisons with certificate storage disabled; and 12 invalid-input checks. A normal Python run and an optimized `python -O` run give identical verification records. The checks use explicit exceptions, so optimization does not remove them.

Table 4 gives five paired timing trials per instance, alternating which implementation runs first. Both retain their exact internal averaging and final-potential checks. Both disable only the serialization of certificate histories for the timing run. The recorded JSON includes all trial times and the actual adjacency matrices, permitting replay without relying on an unspecified input generator.

The benchmark is a small, seeded collection on one execution host, using the recorded Python version and standard-library `Fraction` arithmetic. Five repetitions do not establish a cross-machine performance distribution. The implementation-level saving is nevertheless directly testable: it

Table 4: Measured packet-labeling times. These are comparisons of the existing subroutine and its exact replacement, not full SAT timings.

Rows × columns	Q	Reference (ms)	Cached (ms)	Ratio
12×16	4	1.652	1.100	1.50
16×24	8	10.974	4.111	2.67
20×32	12	39.088	9.140	4.28
24×40	16	103.253	18.320	5.64
24×40	24	304.464	31.833	9.56

performs fewer exact score computations and returns the same objects as the faster relevant prior implementation. It is suitable for integration independently of the proposed new asymptotic schedules.

6 Precision peeling for mode contributions

The source retains a mode tuple β when its guaranteed valuation $V(\beta) = \sum_z w_z \beta_z$ is below the working precision P. It then passes precision P to its child contractions. A further exact transformation removes those guaranteed powers of two before computing the contribution. Its correctness does not require a new Cauchy basis, a particular recursion schedule, or a probabilistic model of the tensor.

6.1 The algebraic identity

Use the source's integral coefficient ring $\mathcal{A} = \mathcal{O}[V]/(V^{\ell})$, with all congruences interpreted coordinatewise. Freeze the current tensor entries at their specified creation-time representatives. Write every targeted mode coefficient as $c_{xz} = 2^{w_z} \bar{c}_{xz}$, where $\bar{c}_{xz} \in \mathcal{A}$. The affine row factors include local terms a_x and the appropriate signs from $l_0(t) = 1 - t$ or $l_1(t) = t$.

Lemma 6.1 (Exact coefficient normalization)**.** *For every multi-index β,*

$$[Z^{\beta}] \prod_x \left(a_x + \sum_z c_{xz} Z_z \right) = 2^{V(\beta)} [Z^{\beta}] \prod_x \left(a_x + \sum_z \bar{c}_{xz} Z_z \right). \tag{31}$$

Proof. The left polynomial is obtained from the right polynomial by substituting $Z_z \mapsto 2^{w_z} Z_z$. Coefficient comparison gives the identity. It remains true with cancellations, nilpotent coordinates, and zero terms. No equality between the valuation of a sum and the minimum valuation of its summands has been assumed. □

Normalize each packet/group tensor by the contribution of the modes it targets, leaving source moments unchanged. Let $v_g(\beta)$ be the sum of target valuations in group g. Multilinearity of contraction gives $Z_{g,\beta} = 2^{v_g(\beta)} \bar{Z}_{g,\beta}$. Every mode has exactly one target, hence $\sum_g v_g(\beta) = V(\beta)$.

Theorem 6.2 (Residual precision)**.** *For every retained tuple define $Q(\beta) = P - V(\beta) > 0$. The recursive mode identity may be evaluated as*

$$F(t + \alpha\zeta) \equiv \sum_{V(\beta) < P} 2^{V(\beta)} \prod_g (\bar{Z}_{g,\beta} \bmod 2^{Q(\beta)}) \pmod{2^P}. \tag{32}$$

The normalized children use the same graph and ports. Tuples with $V(\beta) \geq P$ require no child contraction.

Proof. Apply Lemma 6.1 separately in every packet and group, then substitute in the existing exact factorization. Replacing each normalized factor by its residue modulo 2^Q changes its product by a multiple of 2^Q. The outside factor 2^V makes that error a multiple of 2^P. Addition of contributions preserves the congruence. □

6.2 Division and creation-time representatives

Two details are essential for a valid implementation. First, to obtain $\bar{c}_{xz}$ modulo 2^Q, compute c_{xz} modulo 2^{Q+w_z} and divide its canonical coordinates by 2^{w_z} as ordinary integers. Alternatively, cancel the known powers in the exact factored formula and invert only units. Dividing a residue known only modulo 2^Q would lose w_z output bits. For every used mode, $\beta_z > 0$ implies $w_z \leq V$, so $Q + w_z \leq P$: the original precision already supplies the needed information. Unused modes can be omitted from that coefficient extraction.

Second, residual precision can decrease along a recursive path. The source's original implementation assumes increasing precision and freezes creation-time representatives [3]. Therefore the normalized child must be a newly specified network at precision Q. Reduce each retained parent representative modulo 2^Q and freeze the resulting canonical residue; create normalized packet entries at Q as well. Later lifting uses these new representatives. It must not return to an older, larger representative or reevaluate an unfrozen defining formula.

A retained descriptor can carry a precision cap. Further caps flatten to their minimum, because canonical reductions modulo powers of two compose. This avoids a growing stack of evaluator wrappers. An older entry evaluator may still run at its original creation precision before its result is capped. The theorem does not promise to reduce that cost.

In the cached algorithm, each outer exact-weight array fixes the total V. Consequently $Q = P - V$ is fixed throughout that array, and its entire triangle multiplication can use the same ring $\mathcal{A}/(2^Q)$. Thus peeling is compatible with caching; it does not require adding quantities from different residue rings in a single matrix product.

6.3 What arithmetic work can be saved

Let $N(t)$ count feasible tuples with $V < t$; additional fixed support or degree constraints may be included. The total precision charge for one linear-in-precision operation per contribution has the exact identity

$$\sum_{V(\beta)<P} (P - V(\beta)) = \sum_{t=1}^{P} N(t) \leq PN(P). \tag{33}$$

To prove it, write each positive integer $P - V$ as a sum of $P - V$ ones and exchange the two finite sums. More generally, for nondecreasing f with $f(0) = 0$,

$$\sum_{V(\beta)<P} f(P - V(\beta)) = \sum_{q=1}^{P} (f(q) - f(q-1))N(P - q + 1). \tag{34}$$

These are operation-charge identities, not wall-clock predictions.

There is also a useful certified upper estimate. Put $\mathcal{Z}(\theta) = \prod_z (1 - e^{-\theta w_z})^{-1}$ for positive weights. For any $\alpha > 0$ and $\theta > 0$,

$$\sum_{V(\beta)<P} (P - V(\beta))^{\alpha} \le \left(\frac{\alpha}{e\theta}\right)^{\alpha} e^{\theta P} \mathcal{Z}(\theta). \tag{35}$$

Indeed $q^{\alpha} e^{-\theta q} \le (\alpha/(e\theta))^{\alpha}$ for $q > 0$, and extending the remaining sum to all tuples gives $\mathcal{Z}$. For the same tilted counting certificate, full precision contributes $P^{\alpha} e^{\theta P} \mathcal{Z}(\theta)$. The ratio of these two upper certificates is $(\alpha/(e\theta P))^{\alpha}$. At the contraction saddle, $\theta P \asymp M/\sqrt{u}$, giving a polynomial prefactor saving in this specified arithmetic model. This is not a ratio of actual runtimes, nor a proof that all oracle work scales with residual precision.

An exact dynamic program for four colored families of weights $5 + 4j - 2s_2(j)$ at $P = 1024$ gives mean residual precision about 25.04 bits and linear precision-charge ratio 0.02445. This is an abstract infinite-family partition inventory truncated by the budget, not an assertion that one particular finite packet has that distribution. Finite packet lengths and active valuation levels must still be supplied when using the model for a concrete network.

Peeling leaves the number of mode tuples unchanged. Moreover the existing global schedules make the next Hermite loss dominate the inherited precision. The present argument therefore does not decrease the leading $s/\sqrt{\log s}$ entropy term by itself. Its useful scope is arithmetic that is actually moved to the smaller residue ring.

6.4 Executable evidence

The self-contained prototype checks 544 randomized cyclic tensor contractions, including 120 terms vanishing by valuation. It also compares 18 mode contractions of actual permissible Cauchy subblocks with direct original-kernel sums at precisions from 8 through 80, rejects four invalid normalizations, and runs 1,601 canonical-cap regressions. Normal and optimized Python runs agree. The actual-domain test uses an independent exact rational elimination to reconstruct the relevant Cauchy block, rather than importing the prior feature implementation.

Table 5 measures a separate arithmetic kernel: the cyclic contraction of five dense 12×12 matrices, with input precision $P = 2048$ bits and controlled total divisibility V. Both methods receive the same frozen canonical inputs. The peeled time includes validation and normalization; input generation is excluded from both timings. Seven paired trials alternate execution order.

Table 5: Synthetic cyclic-contraction speed ratios, including normalization. These quantify an arithmetic kernel, not a complete SAT algorithm.

Removed bits V	Residual precision Q	Full/peeled median time
0	2048	0.99
512	1536	1.63
1024	1024	3.11
1536	512	6.63
1792	256	9.48
1984	64	14.34

The zero-valuation row records the small overhead when there is no precision to remove. The other rows show why the transformation can be useful, while leaving open the frequency and cost

share of such operations in a complete implementation. All trial times, seeds, dimensions, and environment metadata are retained in the accompanying JSON.

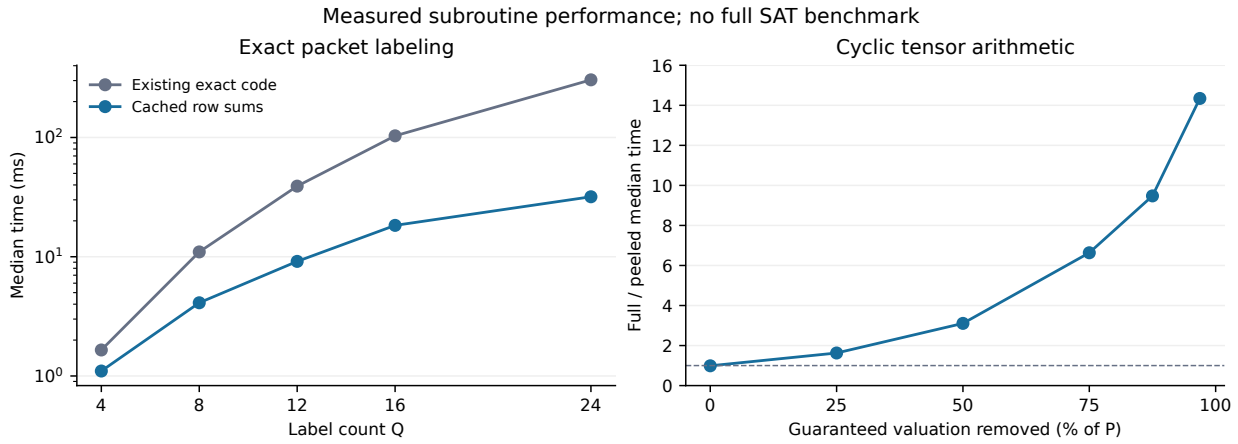


Figure 1: Two distinct measured improvements. Left: the existing exact labeling code versus its cache-preserving replacement. Right: full-precision versus normalized cyclic matrix contraction. The panels have different workloads and should not be multiplied into a predicted SAT speedup.

7 Reproduction, integration, and proof status

7.1 What the archive contains

The report's complete editable source is `report.tex`, with `theory.tex`, `implementation.tex`, `delivery.tex`, and `references.tex`. The compiled `report.pdf` is included. The `code/` directory contains the two implementations, the byte-identical reference labeling routine, exact checkers, benchmark drivers, and a script to regenerate the figure from the recorded JSON. The benchmark results retain raw trials and input data. The mathematical checkers use Python's standard library only; regenerating the figure also requires Matplotlib. Building the PDF requires a standard LaTeX installation with the packages listed in its preamble.

The `provenance/` directory records the checkout commit, relevant commits, source hashes, archive members, and the reference implementation's license. The `notes/` directory retains the detailed review and independent proof-checking notes. Those are research records rather than additional verified theorems. The manifest records every delivered file except the manifest itself. It establishes bundle consistency, not an independent signature or provenance authority.

7.2 Replay

From the extracted archive directory:

```
python3 run_checks.py
python3 code/benchmark_cached_balancing.py --output new_packing_times.json
python3 code/benchmark_precision_peeling.py --output new_precision_times.json
python3 code/make_figures.py
python3 build_report.py
```

On a Windows installation, `py` may replace `python3`. The first command runs all four correctness/parameter checkers normally and with optimization enabled, compares their parsed results, and writes a release-verification summary. The benchmark commands deliberately write new files so the recorded measurements remain intact. Their wall-clock values depend on the execution host and load.

7.3 Integration order

The smallest independently usable change is replacing the asymmetric packet's `balance_pair_labels` implementation with `cached_balancing.py`. It preserves the call signature and returned object. Its exact comparison suite can serve as a migration regression. This does not require adopting a new contraction schedule.

For the polynomial-space mathematical construction, retain the source deformation, group factorization and immutable tensor descriptions. Use asymmetric balancing, the prior full-block Cauchy modes, the exact taper, the buffer (6), and schedule (10). Keep separate degree bounds for retained and fresh sides; replacing them by one maximum would discard the extra capacity used in the proof.

The cached construction is a separate resource option. Use schedule (22), exact directed-pair weight arrays, the specified root/tail block partitions, and recurrence (20). Compute group functions by recursive contraction. Pay for block construction as well as matrix multiplication. Do not quote a cache gain as if the group values were free.

Finally, precision peeling can be applied inside either mode-summation method. Normalize target coefficients before the precision reduction, freeze each new child's capped representatives, and restore the outside power 2^V when accumulating modulo 2^P. The cap regression in the checker shows why retaining an older larger representative during a later lift would give the wrong semantics. This change is appropriate only where those representatives and divisions are handled explicitly.

Claim or component	Evidence and boundary
New multi-stage coefficient below 64	Ordinary proof of gates, precision, recurrence and space; exact rational certificate; independent analytic review. Imported contraction lemmas remain dependencies.
New cached coefficients below 38 and 34	Complete paid recurrence, block/table counts and memory proof; exact interval checks; independent review including the root table term.
Faster exact labeling implementation	Algebraic score identity; exact equality to supplied reference on 840 instances; raw paired benchmark trials. This is an implemented subroutine.
Residual-precision identity	Exact coefficient substitution and congruence proof, valid in the full integral ring; implementation and actual-domain checks over rational scalars. No new extension-ring implementation is supplied.
Full upstream ETH theorem	Imported framework; not independently formalized or rebuilt here. The repository and preprint are identified precisely.

Claim or component	Evidence and boundary
Full fast SAT program or measured SAT acceleration	Not supplied. Local code and timing evidence must not be presented as a full formula-to-count implementation.
Smaller unrestricted logarithmic order	Not established. The missing structured-summation or representation interfaces in the earlier packets remain open.

7.4 Where another substantial gain would need to come from

The delivered changes remove redundant work while preserving established invariants. The direct engineering next step is integrating the exact labeling replacement and residual-precision representation into a complete contraction engine, then measuring their share of actual time. That would answer a question these isolated kernels cannot: whether packing, coefficient extraction, creator-oracle evaluation, or mode enumeration dominates at usable sizes.

For a mathematical order improvement, the concrete missing target is still a mode-summation or compression algorithm with controlled group tensor evaluation cost. The local full-block Smith profile grows linearly, so another invertible integral basis change within that same block cannot make its invariant-factor growth quadratic. That local statement leaves open cancellation across modes, a different deformation, and a more economical SAT-to-network representation. A future claim along any of those routes should specify the actual interface and account for its precision, wire count, oracle cost, and memory before comparing exponents.

A Independent elementary bounds for the cached headline

The interval checker gives many more digits than the theorem needs. For a short independent certificate, let $\xi = \log_2(7/4)$. Exact integer powers give

$$\frac{25}{31} < \xi < \frac{13}{16}, \qquad 7^{31} > 2^{87}, \quad 7^{16} < 2^{45}.$$

The tail ratio satisfies $\gamma < 21/50$, so

$$\frac{a_9}{1-\gamma} < \frac{200}{261} < \frac{\sqrt{566}}{31} < B_9.$$

For the total coefficient, rational brackets from the positive logarithm series and Machin's formula give

$$\kappa < 6.413, \qquad \frac{a_7}{\sqrt{89/5000}} < 3.710, \qquad \frac{(2/7)B_9}{\sqrt{1233/125000}} < 2.213.$$

Their products imply

$$C_{\mathrm{cache}} < 6.413(3.710 + 2.213) = 37.984199 < 38.$$

Similarly $j_7/\sqrt{89/5000} < 5.250$, hence $C_{\mathrm{mem}} < 33.66825 < 34$. All square-root comparisons reduce to integer inequalities after squaring positive quantities. The full independent review also verifies that the first-tail cached-child term exceeds the root table term. The supplied exact interval checker proves all the corresponding strict comparisons directly.

B Source-to-change map

Original source location	Use or replacement
Preprint quantitative contraction: coefficient ring and exact arithmetic	Retain integral extension arithmetic and immutable representative semantics. For peeling, add the explicit creation cap of Section 6.
Preprint packet-packing lemma	Use the incoming asymmetric lemma and preserve its uniform padded-cell arrangement; optionally run the faster exact scorer.
Preprint tapered Hermite recovery	Use the already established exact condition count and six-unit leading precision allowance. The additive buffer remains necessary.
Preprint crossing-mode expansion	Use the joint packet's full-orientation slope-four features, retaining one source and one target group per mode.
Preprint child networks and implicit descriptions	Retain the graph topology and creator descriptors. Track fresh-side degree separately when applying the new schedules.
Joint packet's exact pair-index mode graph	Use its computed group caches and division-free rectangular multiplication, with new alphabets and the full new recurrence.
Asymmetric packet's executable column scoring	Replace repeated row summation by Proposition 5.1; the returned labeling and exact certificate remain identical.

References

[1] Vladimir Reshetnikov and ProveIt contributors. *Sub-exponential 3-SAT project index*, reviewed at commit `474ddba696a5d385c091c23fb2995f69edd13e00`. https://github.com/VladimirReshetnikov/ProveIt/tree/474ddba696a5d385c091c23fb2995f69edd13e00/Computability/SubexponentialSAT.

[2] ProveIt, *New research report(s)*, commit `3e57705cf14ee9ac99c0ac35cfff751dcf70b396`, October 26, 2026. https://github.com/VladimirReshetnikov/ProveIt/commit/3e57705cf14ee9ac99c0ac35cfff751dcf70b396. Seven incoming archives are identified by filename and SHA-256 in the accompanying `provenance/source_manifest.json`. Closing arrival `d098ba436b33df19365a44dd178694ff52845d19` adds the matching packet; placement commit `6d3adc45061535ebe6fa7cd3d9b37dff7dcfc22a` files all eight second-round packets: https://github.com/VladimirReshetnikov/ProveIt/commit/6d3adc45061535ebe6fa7cd3d9b37dff7dcfc22a.

[3] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. Preprint sources pinned to `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`, especially `build/quantitative-contraction.tex` and `build/quantitative-reduction.tex`. https://github.com/openai/math/tree/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a

/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026. The reviewed byte-identical vendored copy is recorded in ProveIt's `lib/openai-math/preprints/PROVENANCE.md`.

[4] *A $2^{o(n)}$ time algorithm for 3-SAT*, credited to Claude (Anthropic), October 26, 2026. The separate 49-page manuscript filed as `Research/claude-2o-n-3sat/eth.pdf`; retrieved from the pinned ProveIt project. Its reported proof development was not supplied for independent execution in this review.

[5] *Cauchy mode improvement*, October 26, 2026. ProveIt `Research/cauchy_mode_improvement/report.tex` and associated code. Integral factorial features and mode-count analysis.

[6] *ETH contraction improvements*, October 26, 2026. ProveIt `Research/eth_contraction_improvements/report.tex`, with schedule and finite-difference extraction artifacts.

[7] *3SAT precision refinement*, October 26, 2026. ProveIt `Research/3sat_precision_refinement/report.tex`. Exact recovery loss, taper minimax calculation, and conditional structured-summation target.

[8] *Density-aware SAT improvement*, October 26, 2026. ProveIt `Research/sat_density_improvement/ density_aware_sat_improvement.tex`.

[9] *ETH density refinement*, October 26, 2026. ProveIt `Research/eth_density_refinement/report.tex`. Density-sensitive bounds and local Cauchy arithmetic.

[10] *3SAT refinement report*, October 26, 2026. ProveIt `Research/3sat_refinement/3sat_refinement_report.tex`. Scalar parity lift and conditional compact recursive factorization.

[11] *3SAT asymptotic analysis*, delivered October 26, 2026, archive name dated October 27. ProveIt `Research/3sat_asymptotic_analysis/report/report.tex`, `research_decision/findings.md`, and companion interface analyses. All seven filed packet paths in references 5–11 are relative to the project directory of reference 1; their immutable full URLs and hashes are in the source manifest.

[12] *Asymmetric packet balancing for subexponential SAT*, October 26, 2026. Incoming archive `eth_asymmetric_packet_balancing_2026-10-26.zip`, especially `report.tex`, `annulus_modes.tex`, and `code/balanced_packing.py`. The delivered reference code has SHA-256 `5e899df4fd25b1371fe6f42c13f87caba49637fb92da0f12b12dd01a831bb6d2`.

[13] *Balanced collision packing for exact counting*, October 26, 2026. Incoming archive `eth_balanced_packing_2026-10-26.zip`; inner directory `eth_balanced_packing`, with `report.tex` and `code/packing.py`.

[14] *High-moment ETH refinement*, October 26, 2026. Incoming archive `eth_high_moment_2026-10-26.zip`; the report's moment potential, orbit padding, and schedule analysis.

[15] *Recursive ETH refinement*, October 26, 2026. Incoming archive `eth_recursive_refinement_2026-10-26.zip`; the report and reciprocal-domain mode analysis.

[16] *SubexponentialSAT review and contraction refinement*, delivered October 26, 2026. Incoming archive `3sat_round2_refinement_20261027.zip`, especially `report/combined.tex`, `report/actual_domain.tex`, and the independent verification notes.

[17] *Sharper packet packing for subexponential exact counting*, October 26, 2026. Incoming archive `eth_packet_refinement_2026-10-26.zip`; inner directory `eth_packet_refinement`, with `report.tex` and code.

[18] *Full crossing-block Cauchy modes*, October 26, 2026. Incoming archive `eth_joint_cauchy_2026-10-26.zip`; `report.tex`, `code/joint_cauchy.py`, `code/mode_graph.py`, and `code/exact_weight_index.py`. The seven original incoming archives in references 12–18 were retrieved at the initial pin `ab2fbc9e03d5c3d0ad6dec8686960bfe498b0b06`. Their relocated files were checked byte-for-byte at the closing pin. Immutable URLs, byte lengths, Git blobs, SHA-256 hashes and member lists accompany this report.

[19] *Sharper quantitative bounds for subexponential SAT contraction: exact matching-based packing and logarithmic precision gaps*, delivered October 26, 2026; report dated October 27. Closing project directory `Research/eth_matching_refinement`, especially `report.tex` and `code/refinement.py`. Originally delivered as `ETH_Matching_Packing_Thin_Gaps_2026-10-27.zip`. https://github.com/VladimirReshetnikov/ProveIt/tree/474ddba696a5d385c091c23fb2995f69edd13e00/Computability/SubexponentialSAT/Research/eth_matching_refinement.